

202510986

Vehicle Registration and Owner Management System (C++)

```
#include <iostream>
#include <fstream>
#include <sstream>
#include <iomanip>
#include <regex>
#include <vector>

using namespace std;

class User {
private:
    string name;
    string email;
    string phone;

public:
    User() {}

    User(string name, string email, string phone) {
        this->name = name;
        this->email = email;
        this->phone = phone;
    }

    string getName() const { return name; }
    string getEmail() const { return email; }
    string getPhone() const { return phone; }

    void setName(string name) { this->name = name; }
    void setEmail(string email) { this->email = email; }
    void setPhone(string phone) { this->phone = phone; }
};

class VehicleOwner : public User {
public:
    VehicleOwner() {}

    VehicleOwner(string name, string email, string phone)
        : User(name, email, phone) {}
};

class Vehicle {
private:
    string plateNumber;
    string vehicleType;
    int year;
    VehicleOwner owner;

public:
```

```

Vehicle() {}

Vehicle(string plateNumber, string vehicleType, int year, VehicleOwner
owner)
    : plateNumber(plateNumber), vehicleType(vehicleType), year(year),
owner(owner) {}

string getPlateNumber() const { return plateNumber; }
string getVehicleType() const { return vehicleType; }
int getYear() const { return year; }
VehicleOwner getOwner() const { return owner; }

string toCSV() const {
    stringstream ss;
    ss << plateNumber << ","
    << vehicleType << ","
    << year << ","
    << owner.getName() << ","
    << owner.getEmail() << ","
    << owner.getPhone();
    return ss.str();
}
};
// Remaining functions: validateEmail, validatePhone, validatePlate,
// addRecord, displayAll, searchVehicle, and main as provided earlier.

```